From Relational Configuration Geometry to a Clock-Free Experimental Benchmark in Hierarchical Shell Manifold Theory

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Abstract

Hierarchical Shell Manifold Theory (HSMT) is a developing framework in which graded relational structure, operator dynamics, and projected effective descriptions are treated as logically distinct layers. A central unresolved issue is that the framework does not presently select a unique fundamental dynamical law. Rather than hiding that underdetermination, the present work turns it into an experimentally discriminable conditional problem.

We identify a minimal finite relational benchmark based on a fixed-margin transportation fiber. For the 2×2 fiber with row and column margins $(2, 2)$, the legal elementary-switch graph is the three-vertex path P_3, whose graph Laplacian has spectrum $\{0, 1, 3\}$. Two previously admissible dynamical completions then give distinct positive modal-rate laws, $\Omega_{\mathrm{S}}(\lambda) = \lambda$ and $\Omega_{\mathrm{I}}(\lambda) = \sqrt{\lambda}$, producing the dimensionless predictions

$$\rho_{\mathrm{S}} = 3, \qquad \rho_{\mathrm{I}} = \sqrt{3}.$$

We show that both models admit a common six-real-state realization, a common initial state, common modal readouts, and a common hysteretic event-counting protocol. The resulting discriminator requires no calibrated external clock: the observable is a ratio of internal event counts over a synchronized interval of complete reference cycles. For the ideal benchmark, two reference periods are sufficient.

We further prove a scalar-copy invariance result for analog implementations. Independent positive scalar gains in the two physical copies of the structural Laplacian and in the two integrator banks combine into a single overall rate scale and therefore cancel from the clock-free ratio. Only nonscalar shape errors affect the spectral discriminator. This observation leads naturally to a non-circular calibration protocol in which structural matrices are measured before dynamical data are examined.

An executable state-space validation and a 20,000-sample component tolerance stress test preserve a wide separation between the two model classes. A complete mixed manufacturer/behavioral SPICE transient and a physical bench experiment remain future work. Consequently this paper establishes an explicit, falsifiable, hardware-realizable conditional benchmark, but makes no claim that either dynamical completion has been selected as the fundamental dynamics of nature.

1 Introduction

Hierarchical Shell Manifold Theory (HSMT) was introduced as a framework for organizing graded relational sectors, operators acting across or within those sectors, and effective dynamics obtained after projection. The framework deliberately separates several notions that are often conflated: hierarchy grade, geometric dimension, spectral dimension, dynamical law, and physical interpretation. In particular, a hierarchy label is not assumed to be a physical spatial dimension, and no identification of a distinguished grade with observed $3 + 1$ spacetime is used in the present work.

The maturation of the framework has also produced an important negative result. The structural assumptions presently available do not uniquely select a master dynamical law. Several internally consistent generators can act on the same positive structural operator. A claim of unique emergent dynamics would therefore be premature.

The purpose of this paper is to turn that underdetermination into a sharply testable conditional question. To do so, we adopt a deliberately chosen finite relational model: the configuration graph of a fixed-margin transportation fiber. This fiber is not derived from the graded operator core of HSMT, nor is it identified with a physical shell or spatial dimension. It is used solely because its elementary-switch structure can be stated exactly and because it supplies the smallest configuration graph whose Laplacian has two positive eigenvalues, the minimum required for an internal modal-rate comparison.

On this model the central chain is

$$\boxed{\begin{aligned}&\text{relational configuration structure} \longrightarrow K \longrightarrow \text{conditional dynamics}\\ &\qquad\qquad\qquad\qquad\qquad\longrightarrow \text{modal-rate ratio} \longrightarrow \text{internal event record.}\end{aligned}}$$

Two admissible dynamical completions of the same structural operator then produce distinct dimensionless predictions. The comparison is clock-free in the operational sense relevant here: no externally calibrated time unit is required. Only the ratio of internally generated periodic event counts is used. Both completions can be realized in the same six-real-state analog architecture, with the same structural operator, the same preparation, the same readouts, and the same event detector; only one feedback map changes.

This paper therefore has a deliberately narrow goal. It does not attempt to derive observed spacetime, cosmology, gravitation, gauge structure, or a unique fundamental dynamics. It establishes instead that a nontrivial conditional sector compatible with the present underdetermination of HSMT has become mathematically explicit enough to support a falsifiable experimental benchmark.

2 Scope and claim boundary

The analysis rests on a strict separation between structural, dynamical, operational, and physical claims.

The structural object developed below is a finite graph Laplacian arising from a transportation fiber. The graph is relational. It is not identified with physical space.

The two dynamical models are conditional completions of the same structural operator. Neither is claimed to be selected by the present HSMT axioms.

The experimental realization is an emulator of the conditional mathematics. Successful realization would show that the proposed relational-to-operational chain can be instantiated physically. It would not by itself establish HSMT as a fundamental theory.

Throughout the paper we therefore maintain the following firewall:

$$\boxed{\text{emulator validation} \neq \text{fundamental-physics validation.}}$$

In particular, nothing in the present work establishes a physical identification of a hierarchy grade N, a Gaussian shell profile, a physical shell width, physical $3+1$ dimensionality, or a cosmological interpretation.

3 Fixed-margin transportation fibers

Let

$$\rho = (\rho_1, \dots, \rho_m), \qquad \kappa = (\kappa_1, \dots, \kappa_n)$$

be nonnegative integer row and column margins satisfying

$$\sum_{i=1}^{m} \rho_i = \sum_{j=1}^{n} \kappa_j.$$

The corresponding transportation fiber is

$$\mathcal{T}(\rho, \kappa) = \left\{ T \in \mathbb{Z}_{\geq 0}^{m \times n} : T\mathbf{1} = \rho, \quad T^T\mathbf{1} = \kappa \right\}.$$

The elementary margin-preserving move is the 2×2 rectangle switch

$$\Delta_{ab;cd} = E_{ac} + E_{bd} - E_{ad} - E_{bc},$$

where $a \neq b$ and $c \neq d$.
Whenever both

$$T + \Delta_{ab;cd}$$

and

$$T - \Delta_{ab;cd}$$

remain nonnegative, the corresponding configurations are adjacent in the elementary-switch graph.

For ordinary two-way nonnegative integer transportation fibers, these rectangle moves form the standard local algebraic mechanism preserving the margins. Their role in the present paper is purely structural: they define the finite configuration graph on which the benchmark operator acts.

4 The minimal three-state benchmark

Consider the 2×2 equal-margin family

$$\rho = \kappa = (r, r), \qquad r \in \mathbb{Z}_{\geq 1}.$$

Every table is uniquely determined by one integer a:

$$T(a) = \begin{pmatrix} a & r-a \\ r-a & a \end{pmatrix}, \qquad a = 0, 1, \dots, r.$$

Hence the fiber contains

$$r + 1$$

states.
The single elementary rectangle move sends

$$a \mapsto a \pm 1,$$

whenever the resulting table remains nonnegative.

Therefore the configuration graph is exactly

$$P_{r+1},$$

the path graph on $r+1$ vertices.

Proposition 1. *For the 2×2 equal-margin fiber with $\rho = \kappa = (r, r)$, the elementary-switch graph is the path P_{r+1}.*

Proof. The margin constraints force the table to have the form $T(a)$ above. The only nontrivial margin-preserving rectangle move increments or decrements a by one. Therefore each interior state is adjacent to its two nearest neighbors in a, and each endpoint has one neighbor. This is precisely P_{r+1}. □

For $r = 1$, the graph is P_2, whose Laplacian has one zero eigenvalue and only one positive eigenvalue. No nontrivial internal positive-mode ratio can therefore be formed.

The next case,

$$r = 2,$$

contains the three states

$$T_0 = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}, \qquad T_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad T_2 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

Their graph is

$$T_0 \longleftrightarrow T_1 \longleftrightarrow T_2.$$

Thus the minimal benchmark structural operator is the unit-weight P_3 graph Laplacian

$$K = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

Its characteristic polynomial is

$$\det(K - \lambda I) = -\lambda(\lambda - 1)(\lambda - 3),$$

so

$$\boxed{\operatorname{spec}(K) = \{0, 1, 3\}.}$$

A convenient orthonormal eigenbasis is

$$\phi_0 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},$$

$$\phi_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix},$$

and

$$\phi_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix},$$

with

$$K\phi_0 = 0, \qquad K\phi_1 = \phi_1, \qquad K\phi_2 = 3\phi_2.$$

Theorem 1 (Minimal positive-mode benchmark)**.** *A finite connected graph requires at least three vertices in order for its Laplacian to possess two positive eigenvalues. The fiber $\mathcal{T}((2,2),(2,2))$ realizes this minimum with exactly three configuration states.*

Proof. The Laplacian of a connected graph on n vertices has one zero eigenvalue and $n-1$ positive eigenvalues. Two positive eigenvalues therefore require $n \geq 3$. The fiber above has exactly three states and hence saturates the bound. □

The benchmark is therefore minimal by state count.

5 Two conditional dynamical completions

The structural operator K does not, by itself, determine a unique dynamical law. We therefore consider two explicitly declared conditional completions.

5.1 S completion

The first completion is

$$\dot{z} = -igKz,$$

where

$$z \in \mathbb{C}^3$$

and

$$g > 0$$

sets the overall laboratory rate scale.
For a K-eigenvector with eigenvalue λ_j,

$$z_j(t) = e^{-ig\lambda_j t} z_j(0),$$

so the positive modal rates are

$$\boxed{\Omega_j^{(\mathrm{S})} = g\lambda_j.}$$

For the benchmark eigenvalues 1 and 3,

$$\Omega_1^{(\mathrm{S})} = g, \qquad \Omega_2^{(\mathrm{S})} = 3g.$$

Hence

$$\boxed{\rho_{\mathrm{S}} = \frac{\Omega_2^{(\mathrm{S})}}{\Omega_1^{(\mathrm{S})}} = 3.}$$

5.2 I completion

The second completion is the positive-sector wave equation

$$\ddot{u} + g^2 K u = 0.$$

On a K-eigenvector,

$$\ddot{u}_j + g^2 \lambda_j u_j = 0,$$

so

$$\boxed{\Omega_j^{(\mathrm{I})} = g\sqrt{\lambda_j}.}$$

Thus

$$\Omega_1^{(\mathrm{I})} = g, \qquad \Omega_2^{(\mathrm{I})} = \sqrt{3}\, g,$$

and

$$\boxed{\rho_{\mathrm{I}} = \sqrt{3}.}$$

The same structural spectrum therefore produces the sharply distinct dimensionless predictions

$$\boxed{3 \quad \text{versus} \quad \sqrt{3}.}$$

No absolute calibration of g is required to distinguish them.

6 Clock-free discrimination

The phrase "clock free" is used here in a precise operational sense. The physical device evolves in laboratory time, but the discriminating observable does not require the laboratory second or any other external time unit to be calibrated.

Let a reference mode generate periodic events at rate Ω_1, and let a test mode generate events at rate Ω_2.

The relevant quantity is

$$\rho = \frac{\Omega_2}{\Omega_1}.$$

If the observation interval is defined internally by exactly M complete reference periods, the test count N_2 gives

$$\widehat{\rho} = \frac{N_2}{M}.$$

A change of laboratory clock scale

$$t \mapsto ct$$

multiplies both rates by $1/c$ and therefore leaves ρ unchanged.

This is the operational reason that the discriminator is independent of an absolute clock calibration.

7 A common six-real-state realization

The two conditional dynamics can be implemented in the same six-dimensional real state space.

Let

$$z = u + iv, \qquad u, v \in \mathbb{R}^3.$$

For the S completion,

$$\dot{z} = -igKz$$

is equivalent to

$$\boxed{\dot{u} = gKv, \qquad \dot{v} = -gKu.}$$

Hence

$$\dot{Y} = A_{\mathrm{S}} Y, \qquad Y = \begin{pmatrix} u \\ v \end{pmatrix},$$

with

$$A_{\mathrm{S}} = g \begin{pmatrix} 0 & K \\ -K & 0 \end{pmatrix}.$$

Because $K^T = K$,

$$A_{\mathrm{S}}^T = -A_{\mathrm{S}},$$

and therefore

$$\frac{d}{dt}\left(\|u\|^2 + \|v\|^2\right) = 0.$$

For the I completion, define

$$v = \frac{1}{g}\dot{u}.$$

Then

$$\ddot{u} + g^2 K u = 0$$

becomes

$$\boxed{\dot{u} = gv, \qquad \dot{v} = -gKu.}$$

Thus

$$A_{\mathrm{I}} = g \begin{pmatrix} 0 & I \\ -K & 0 \end{pmatrix}.$$

The corresponding conserved quadratic form on the positive spectral sector is

$$E_{\mathrm{I}} = \frac{1}{2}\left(u^T K u + v^T v\right).$$

The same real hardware state space can therefore realize both models.
Only the upper-right feedback block changes:

$$K \quad \longleftrightarrow \quad I.$$

8 Common preparation and readouts

A particularly convenient common initial state is

$$\boxed{u(0) = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \qquad v(0) = 0.}$$

Because the entries of $u(0)$ sum to zero,

$$\phi_0^T u(0) = 0,$$

so the zero mode is absent.
The positive-mode amplitudes are

$$\phi_1^T u(0) = \frac{1}{\sqrt{2}},$$

and

$$\phi_2^T u(0) = \frac{\sqrt{6}}{2},$$

both nonzero.
Define normalized readouts

$$\boxed{X_1 = u_1 - u_3}$$

and

$$\boxed{X_2 = \frac{u_1 - 2u_2 + u_3}{3}.}$$

These are normalized versions of the projections onto ϕ_1 and ϕ_2.
For the S dynamics,

$$\boxed{X_1^{(\mathrm{S})}(t) = \cos(gt),}$$

$$\boxed{X_2^{(\mathrm{S})}(t) = \cos(3gt).}$$

For the I dynamics,

$$\boxed{X_1^{(\mathrm{I})}(t) = \cos(gt),}$$

$$\boxed{X_2^{(\mathrm{I})}(t) = \cos(\sqrt{3}\, gt).}$$

The reference channel is therefore exactly common:

$$\boxed{X_1^{(\mathrm{S})} = X_1^{(\mathrm{I})}.}$$

This provides an internal experimental control.

A substantial change in the measured reference behavior between S and I would indicate a hardware or calibration problem rather than the intended model distinction.

9 Hysteretic event detection

Frequency estimation need not be performed by Fourier analysis.

For each normalized readout $X(t)$, define two thresholds

$$-h \qquad \text{and} \qquad +h,$$

with

$$\boxed{h = \frac{1}{2}.}$$

The detector has two states.

In the high state, it waits until

$$X \leq -h$$

and then enters the low state.

In the low state, it waits until

$$X \geq +h$$

and then returns to the high state while registering one completed event.

Thus one full ordered threshold excursion produces exactly one count.

For an ideal unit-amplitude sinusoid, this yields one event per period.

If the observed signal has the form

$$X(t) = A\cos(\Omega t + \varphi) + \eta(t),$$

a sufficient robustness condition is

$$\|\eta\|_\infty < h$$

and

$$A > h + \|\eta\|_\infty.$$

For $A = 1$ and $h = 1/2$, continuous additive contamination smaller than one half in supremum norm cannot create an additional complete ordered hysteretic event.

10 The synchronized observation window

A subtle endpoint issue arises if event counters are simply started at an arbitrary phase and stopped after M completed reference events.

That procedure does not necessarily span exactly M complete reference periods.

The formal protocol therefore uses reference-event synchronization.

Let

$$r_1, r_2, r_3, \ldots$$

denote successive completed reference events.

The first completed event r_1 is used only for synchronization.

The formal observation interval for M reference periods is

$$\boxed{(r_1, r_{M+1}].}$$

Exactly M complete reference periods lie inside this interval.

If N_2 test events occur there, define

$$\boxed{\widehat{\rho} = \frac{N_2}{M}.}$$

Proposition 2 (Finite event-count bound)**.** *For an exactly periodic test event sequence observed over an interval of M complete reference periods,*

$$|\widehat{\rho} - \rho| \leq \frac{1}{M}.$$

Proof. Over a reference-defined interval of duration T, the real-valued expected number of test periods is ρM. The integer number of completed periodic events in the same interval differs from this number by at most one due to endpoint placement. Therefore

$$|N_2 - \rho M| \leq 1.$$

Dividing by M gives the result. □

Consequently two candidate ratios are guaranteed to have disjoint finite-count bands if

$$\boxed{|\rho_A - \rho_B| > \frac{2}{M}.}$$

For the present benchmark,

$$3 - \sqrt{3} \approx 1.267949.$$

For

$$M = 2,$$

$$\frac{2}{M} = 1,$$

so

$$1 < 3 - \sqrt{3}.$$

Thus

$$\boxed{M = 2}$$

is sufficient.

For the ideal synchronized window the S model produces

$$\boxed{N_2^{(\mathrm{S})} = 6,}$$

so

$$\widehat{\rho}_{\mathrm{S}} = 3.$$

The I model produces

$$\boxed{N_2^{(\mathrm{I})} = 4,}$$

so

$$\widehat{\rho}_{\mathrm{I}} = 2.$$

The latter is not equal to the asymptotic modal ratio $\sqrt{3}$, but it satisfies

$$|2 - \sqrt{3}| < \frac{1}{2}.$$

The event record therefore discriminates the two models without requiring a calibrated external time unit.

11 Scalar-copy invariance

A physical analog device does not contain an abstract matrix K. It contains coefficient networks and integrators with finite component tolerances.

A useful structural invariance substantially reduces the resulting calibration problem.

Let the lower physical structural block be

$$K_A = aK,$$

and the upper S block be

$$K_B = bK,$$

where

$$a, b > 0.$$

Let the I upper branch be

$$H_B = bI.$$

Let the three u-integrators share a common gain g_u, and the three v-integrators share a common gain g_v:

$$G_u = g_u I, \qquad G_v = g_v I.$$

No equality between a and b, or between g_u and g_v, is required.

Theorem 2 (Scalar-copy invariance)**.** *Under*

$$K_A = aK, \qquad K_B = bK, \qquad H_B = bI,$$

and

$$G_u = g_u I, \qquad G_v = g_v I,$$

all positive scalar gains combine into a single overall rate

$$\boxed{c = \sqrt{g_u g_v ab}.}$$

The S and I modal-rate ratios remain exactly

$$\boxed{\rho_{\mathrm{S}} = 3, \qquad \rho_{\mathrm{I}} = \sqrt{3}.}$$

Proof. The physical S system is

$$\dot{u} = g_u bKv, \qquad \dot{v} = -g_v aKu.$$

Therefore

$$\ddot{u} = -g_u g_v abK^2 u = -c^2 K^2 u.$$

For

$$K\phi_j = \lambda_j \phi_j,$$

the S modal rates are

$$\Omega_j^{(\mathrm{S})} = c\lambda_j.$$

Hence

$$\frac{\Omega_2^{(\mathrm{S})}}{\Omega_1^{(\mathrm{S})}} = \frac{\lambda_2}{\lambda_1} = 3.$$

For I,

$$\dot{u} = g_u bv, \qquad \dot{v} = -g_v aKu,$$

so

$$\ddot{u} = -g_u g_v abKu = -c^2 Ku.$$

Therefore

$$\Omega_j^{(\mathrm{I})} = c\sqrt{\lambda_j},$$

and

$$\frac{\Omega_2^{(\mathrm{I})}}{\Omega_1^{(\mathrm{I})}} = \sqrt{\frac{\lambda_2}{\lambda_1}} = \sqrt{3}.$$

□

This theorem identifies a useful hardware distinction:

$$\boxed{\text{scalar gain error} \quad \text{versus} \quad \text{shape error.}}$$

Scalar errors merely alter the overall laboratory rate.
Nonscalar errors alter the relative modal structure and must therefore be measured or bounded.

12 General calibrated hardware model

For a nonideal board, let

$$G_u = \mathrm{diag}(g_{u1}, g_{u2}, g_{u3}),$$

$$G_v = \mathrm{diag}(g_{v1}, g_{v2}, g_{v3}),$$

and let the measured physical transfer matrices be

$$K_A, \qquad K_B, \qquad H_B.$$

Then the calibrated S state matrix is

$$\boxed{A_{\mathrm{S}}^{\mathrm{cal}} = \begin{pmatrix} 0 & G_u K_B \\ -G_v K_A & 0 \end{pmatrix},}$$

while the calibrated I state matrix is

$$\boxed{A_{\mathrm{I}}^{\mathrm{cal}} = \begin{pmatrix} 0 & G_u H_B \\ -G_v K_A & 0 \end{pmatrix}.}$$

These matrices must be determined before the transient dynamical record is examined.
This produces an explicit anti-circularity requirement:

$$\boxed{\text{static calibration} \longrightarrow \text{predicted dynamics} \longrightarrow \text{transient comparison},}$$

not

$$\boxed{\text{transient data} \longrightarrow \text{redefinition of structure.}}$$

13 Static calibration protocol

Each integrator gain can be measured by disconnecting the closed dynamical loop and applying equal positive and negative test voltages.

If the measured ramp slopes are

$$s_i^+$$

for $+V_T$ and

$$s_i^-$$

for $-V_T$, then

$$\boxed{g_i = \frac{s_i^- - s_i^+}{2V_T}.}$$

The symmetric measurement suppresses first-order constant offset drift.

For each physical matrix block, inject the three Cartesian basis vectors with both signs.

If

$$y_j^+$$

and

$$y_j^-$$

are the outputs produced by

$$+V_T e_j$$

and

$$-V_T e_j,$$

respectively, then the measured matrix column is

$$\boxed{(K_{\text{meas}})_{\cdot j} = \frac{y_j^+ - y_j^-}{2V_T}.}$$

This procedure determines K_A, K_B, and H_B without using oscillation frequencies.

For diagnostic purposes, the best scalar fit of a measured block K_A to the ideal structure K in Frobenius norm is

$$a = \frac{\langle K_A, K \rangle_F}{\|K\|_F^2},$$

with

$$\langle A, B \rangle_F = \operatorname{tr}(A^T B).$$

The residual

$$R_A = K_A - aK$$

measures nonscalar shape error.

Analogous definitions apply to K_B.

This decomposition is useful because a large difference between the scalar fits a and b is not itself a failure. The relevant distortion is contained in the residual matrices.

14 Predeclared acceptance criterion

Suppose static calibration produces predicted modal ratios

$$\rho_{\mathrm{S}}^{\mathrm{cal}}$$

and

$$\rho_{\mathrm{I}}^{\mathrm{cal}},$$

with calibration uncertainties

$$\delta\rho_{\mathrm{S}}, \qquad \delta\rho_{\mathrm{I}}.$$

A sufficient condition for disjoint finite-count prediction intervals is

$$\boxed{\left|\rho_{\mathrm{S}}^{\mathrm{cal}} - \rho_{\mathrm{I}}^{\mathrm{cal}}\right| > \frac{2}{M} + \delta\rho_{\mathrm{S}} + \delta\rho_{\mathrm{I}}.}$$

If the calibrated separation is

$$\Delta_{\mathrm{cal}} = \left|\rho_{\mathrm{S}}^{\mathrm{cal}} - \rho_{\mathrm{I}}^{\mathrm{cal}}\right|$$

and

$$\delta_{\mathrm{cal}} = \delta\rho_{\mathrm{S}} + \delta\rho_{\mathrm{I}},$$

then any integer satisfying

$$\boxed{M > \frac{2}{\Delta_{\mathrm{cal}} - \delta_{\mathrm{cal}}}}$$

is sufficient whenever

$$\Delta_{\mathrm{cal}} > \delta_{\mathrm{cal}}.$$

Crucially, M must be chosen before the transient classification record is inspected.

15 Executable state-space validation

The ideal six-state system has been evaluated numerically as an executable continuous-time state-space model.

For

$$g = 100 \text{ s}^{-1},$$

the S positive angular frequencies are

$$100 \text{ rad/s}$$

and

$$300 \text{ rad/s}.$$

The corresponding ordinary frequencies are approximately

$$15.9155 \text{ Hz}$$

and

$$47.7465 \text{ Hz}.$$

The I positive angular frequencies are

$$100 \text{ rad/s}$$

and

$$100\sqrt{3} \text{ rad/s},$$

corresponding to approximately

$$15.9155 \text{ Hz}$$

and

$$27.5664 \text{ Hz}.$$

The numerical trajectories reproduce

$$X_1(t) = \cos(100t)$$

and the expected S and I test channels to floating-point accuracy.

A deliberately nontrivial scalar-copy test using

$$a = 1.07, \qquad b = 0.93,$$

$$g_u = 101.3 \text{ s}^{-1}, \qquad g_v = 98.7 \text{ s}^{-1}$$

gives the common rate scale

$$c = \sqrt{g_u g_v a b} \approx 99.74627 \text{ rad/s}.$$

The resulting numerical ratios remain exactly

$$3$$

and

$$\sqrt{3}$$

up to numerical precision, confirming the scalar-copy theorem in the executable model.

The state-space calculation also revealed the event-window issue described above: starting counters at an arbitrary initial phase and stopping after the second completed reference event does not generally span two complete reference periods. This motivated the synchronized event-window protocol used throughout the present paper.

16 Component-tolerance stress test

A separate 20,000-sample nonideal linear tolerance ensemble was used as an engineering stress test.

The sampled perturbation ranges included approximately:

$$\pm 0.1\%$$

integrator-resistor variation,

$$\pm 1\%$$

integrator-capacitor variation,

$$\pm 0.1\%$$

coefficient perturbations in the two physical structural blocks,
small independent gain perturbations in the I branch,
and analog-switch series resistance spanning the relevant sub-50 Ω range.
No post-sample dynamical fitting was used.
The resulting S ratio range was

$$\boxed{2.96588 \leq \rho_{\mathrm{S}} \leq 3.03317.}$$

The I ratio range was

$$\boxed{1.71171 \leq \rho_{\mathrm{I}} \leq 1.75170.}$$

The minimum observed separation between the two ensembles was

$$\boxed{\Delta_{\min} \approx 1.25382.}$$

For comparison, the ideal separation is

$$3 - \sqrt{3} \approx 1.26795.$$

Thus no sampled realization approached overlap.

This result should not be interpreted as a statistical confidence interval for manufactured hardware. It is a bounded engineering stress test demonstrating that the ideal discrimination gap is large compared with representative component-scale perturbations.

For a final reproducibility archive, the exact sampling distribution, software version, random-number seed, and source code should accompany the manuscript.

17 Analog implementation

A straightforward analog realization uses six inverting integrators for the state variables

$$u_1, u_2, u_3, v_1, v_2, v_3.$$

With nominal values

$$R_I = 100\,\mathrm{k}\Omega, \qquad C_I = 100\,\mathrm{nF},$$

the nominal scale is

$$g = \frac{1}{R_I C_I} = 100\ \mathrm{s}^{-1}.$$

The P_3 Laplacian can be implemented without a general matrix multiplier.

For any vector

$$x = (x_1, x_2, x_3)^T,$$

define

$$q_1 = x_1 - x_2,$$

$$q_3 = x_3 - x_2.$$

Then

$$q_2 = -(q_1 + q_3)$$

and

$$\begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = Kx.$$

Thus each physical K block requires only two difference stages and one inverting summer.

The S/I selector changes only the upper feedback branch:

$$K_B v \quad \longleftrightarrow \quad H_B v,$$

while the lower branch

$$-K_A u$$

remains fixed.

The same physical preparation, readouts, thresholds, and event logic are retained in both runs.

A pin-level Rev-A design has been specified using precision quad operational amplifiers, analog switches for reset and mode routing, dual comparators, digital S-R latches, and event counters. That design is an engineering realization of the equations above rather than an additional theoretical assumption.

18 Device-level readiness and remaining SPICE work

The device design has undergone a datasheet-bounded nonideality audit.

At the intended tens-of-hertz operating frequencies, representative precision operational-amplifier bandwidth and slew-rate limits are many orders of magnitude above the required values.

Analog-switch on resistance contributes a sub-per-mille correction relative to the nominal 100 kΩ integrator resistance.

Comparator propagation delay is negligible relative to the millisecond-scale oscillation periods.

The largest remaining device-level drift mechanism identified in the current design audit is reset-switch off-state leakage acting on the 100 nF integrating capacitors. The resulting worst-case datasheet-scale drift remains small compared with the 1-V modal amplitude and 0.5-V hysteretic threshold, but it merits direct device-level simulation and later bench measurement.

At the time of this manuscript draft, the full mixed-device transient simulation has not been executed.

The intended mixed simulation stack consists of:

manufacturer OPAx197-family model,

ADG333A switch model,

ADG1412 reset-switch model,

and

a datasheet-bounded behavioral LM319 comparator model.

The mixed-model simulation must reproduce the same anti-circular sequence required of the physical experiment:

static calibration → precomputed prediction → transient → EW-0 event extraction.

No claim of a full manufacturer-macromodel SPICE pass is made in this version of the paper.

19 Falsifiability

The present construction is falsifiable at several distinct levels.

A realization fails the benchmark if the independently calibrated structural matrices do not support the predicted oscillatory sectors.

It fails if the common reference channel differs materially between S and I beyond the calibration uncertainty.

It fails if the readout contamination violates the hysteretic mode-purity criterion.

It fails if the calibrated S and I prediction intervals overlap at the predeclared observation length.

It fails if the synchronized event records disagree with the precomputed conditional predictions.

Most importantly, the structural parameters may not be repaired using the same transient record they are supposed to predict.

This prevents the benchmark from becoming an unconstrained fitting exercise.

20 Interpretation

What would a successful benchmark establish?

It would establish that a finite relational configuration structure derived from fixed-margin transportation data can be mapped to a physical dynamical emulator in which two conditional HSMT-compatible dynamical completions make sharply different, internally measurable predictions.

It would validate the operational chain

$$\boxed{\text{relational configuration graph} \to K \to \text{conditional dynamics} \to \text{modal rates} \to \text{hysteretic event record.}}$$

It would also validate the practical utility of clock-free modal ratios and the scalar-copy invariance of the hardware realization.

It would not establish that the transportation graph is physical space, that either conditional dynamical law is fundamental, or that the broader HSMT hierarchy has been uniquely realized in nature.

The experiment is therefore best understood as a bridge test between a mathematical HSMT sector and controlled physical implementation.

21 Relation to the broader HSMT program

The present result occupies a specific place in the larger HSMT architecture.

Earlier work introduced the graded arena and operator framework, with schematically distinct stages

$$\mathcal{M} = \bigsqcup_N \mathcal{M}_N \longrightarrow \text{operators} \longrightarrow \text{states} \longrightarrow \text{projection} \longrightarrow H_{\text{eff}}.$$

Subsequent analysis showed that structural information alone does not select a unique generator. The present paper does not remove that underdetermination.

Instead, it establishes that conditional dynamical alternatives can be made operationally distinguishable before the ultimate foundational selector is known.

This is an important methodological change.

Rather than requiring the entire theory to be closed before any test is possible, one may isolate a conditional sector, derive dimensionless observables that are robust to unphysical scale choices, and test the result independently.

In this sense the present benchmark does not complete HSMT.

It makes part of HSMT experimentally addressable.

22 Limitations

Several limitations should remain explicit.

First, the transportation-fiber construction is relational and is not identified with observed physical space.

Second, the two dynamics tested here are conditional alternatives. The present framework does not derive which, if either, is realized by nature.

Third, the analog circuit is an emulator. Its successful operation would validate a mathematical-to-operational mapping, not establish the ontology of the underlying universe.

Fourth, a complete mixed-device SPICE transient has not yet been executed in this manuscript version.

Fifth, no physical bench result is reported here.

Sixth, no physical identification is made between the present three-state benchmark and a hierarchy grade N.

Seventh, no Gaussian shell profile or shell width enters the benchmark.

Eighth, no cosmological conclusion is drawn.

These restrictions are part of the result rather than weaknesses to be hidden. They define exactly what evidence the proposed benchmark can and cannot supply.

23 Conclusion

We have constructed a minimal clock-free experimental benchmark for a conditional relational sector of Hierarchical Shell Manifold Theory.

The benchmark begins with the 2×2 transportation fiber with equal margins $(2, 2)$. Its three configurations form the path graph P_3, whose Laplacian spectrum is

$$\{0, 1, 3\}.$$

Two admissible dynamical completions produce the exact modal-rate predictions

$$\boxed{\rho_{\mathrm{S}} = 3, \qquad \rho_{\mathrm{I}} = \sqrt{3}.}$$

Both can be implemented in the same six-real-state device with a common structural operator, common preparation, common reference mode, common readouts, and common event detector.

A synchronized hysteretic observation over two complete reference periods is sufficient for ideal discrimination.

The physical realization also possesses a useful scalar-copy invariance: absolute positive gain differences between structural blocks and integrator banks combine into one unobservable clock scale, while only nonscalar shape errors affect the dimensionless ratio.

This leads to a non-circular calibration protocol in which the complete hardware state matrices are determined before transient data are examined.

Executable state-space calculations and a 20,000-sample component tolerance stress test show a large separation margin between the two conditional predictions.

The next evidential step is no longer another internal derivation. It is an actual mixed-model circuit simulation followed by physical bench measurement.

Accordingly, the present work marks a transition in HSMT from purely structural development to a predeclared experimental benchmark.

A General P_{r+1} spectrum

For the path graph P_{r+1}, the Laplacian eigenvalues are

$$\lambda_j = 2 - 2\cos\left(\frac{\pi j}{r+1}\right), \qquad j = 0, 1, \ldots, r.$$

For $r = 2$,

$$\lambda_0 = 0,$$

$$\lambda_1 = 2 - 2\cos\left(\frac{\pi}{3}\right) = 1,$$

and

$$\lambda_2 = 2 - 2\cos\left(\frac{2\pi}{3}\right) = 3.$$

For $r = 3$, the next benchmark is P_4, with

$$\lambda_0 = 0,$$

$$\lambda_1 = 2 - \sqrt{2},$$

$$\lambda_2 = 2,$$

and

$$\lambda_3 = 2 + \sqrt{2}.$$

The P_4 system therefore supplies three positive modes and can be used for overdetermined modal-law consistency tests.

B Loop consistency for larger benchmarks

Suppose a model assigns one positive rate

$$\Omega(\lambda_j)$$

to each positive structural eigenvalue.

For three modes define

$$\rho_{21} = \frac{\Omega_2}{\Omega_1},$$

$$\rho_{32} = \frac{\Omega_3}{\Omega_2},$$

and

$$\rho_{31} = \frac{\Omega_3}{\Omega_1}.$$

Any consistent single-valued rate law must satisfy

$$\boxed{\rho_{21}\rho_{32} = \rho_{31}.}$$

Thus the P_4 extension tests more than S versus I. It begins to test the assumption that one stable modal rate is associated with each structural eigenvalue.

C General nonscalar hardware equations

For the measured physical matrices

$$G_u, \quad G_v, \quad K_A, \quad K_B, \quad H_B,$$

the S system is

$$\dot{u} = G_u K_B v,$$

$$\dot{v} = -G_v K_A u,$$

and therefore

$$\boxed{\ddot{u} = -G_u K_B G_v K_A u.}$$

The I system is

$$\dot{u} = G_u H_B v,$$

$$\dot{v} = -G_v K_A u,$$

so

$$\boxed{\ddot{u} = -G_u H_B G_v K_A u.}$$

These equations show explicitly why independent calibration of the physical blocks is necessary once nonscalar imperfections are present.

D Suggested experimental record

A future physical experiment should archive, at minimum,

$$G_u, \qquad G_v, \qquad K_A, \qquad K_B, \qquad H_B,$$

the threshold voltages,
the supply rails,
the preparation state,
the S and I X_1 and X_2 traces,
the reference and test event streams,
the selected observation length M,
and the complete component revision.
The structural quantities must be established before the dynamical record is used for comparison.

E Reproducibility status

The analytical benchmark is fully specified by the matrices and equations in the main text.

The executable state-space validation should be archived with source code.

The 20,000-sample tolerance study should, prior to submission, be supplemented with:

- the exact random-number seed;
- the exact probability distribution used for every tolerance;
- the numerical eigensolver and software version;
- the complete sampling code;
- machine-readable trial results or a compact reproducibility archive.

The mixed-model SPICE archive should likewise include the precise manufacturer model revisions and any syntax-only compatibility edits.

Acknowledgment of epistemic status

The benchmark developed here is intentionally conditional. Its value lies in producing a sharply discriminable experimental consequence without claiming that the unresolved foundational selection problem has already been solved.

A future physical result should therefore be reported whether it agrees with, partially agrees with, or contradicts the benchmark prediction.

References

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